Lab 4: TokTickIT — Actions Taken, Dashboards, and Final Regression

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Answer Part 1: Git Use with Engineering Workflow

GitHub Repository URL: https://github.com/Decaa0/toktickit

Every feature was isolated in a dedicated branch, peer-reviewed via pull request, approved by the peer reviewer (Punnapob Wirojwongchai — SaintCrois), and merged into `lab4-staging` before final integration into `main`.

Issue	Scope	Pull Request
#44	Issue 4.1: Sprint 4 Engineering Contract & Specifications	#56 — feature/4.1-sprint4-specs (Approved)
#45	Issue 4.2: Actions Taken foundation (model, migration, seed, API)	#57 — feature/4.2-actions-taken-foundation (Approved)
#47	Issue 4.4: Ticket workflow & resolution gate	#58 — feature/4.4-ticket-workflow (Approved)
#48	Issue 4.5: Role dashboards (Requester & IT Staff)	#59 — feature/4.5-role-dashboards (Approved)
#46	Issue 4.3: Actions Taken UI	#60 — feature/4.3-actions-taken-ui (Approved)
#49	Issue 4.6: Responsive fix & final hardening	#61 — feature/4.6-hardening-responsive (Approved)

[Git History Screenshot — feature branches → lab4-staging → main]

[Repository Directory Structure Screenshot]

Rendered reviewer record: `docs/lab-04/reviewer.md`. README, `.gitignore`, and the repository structure are current on `main`.

Answer Part 2: Spec DD

Rendered: `docs/lab-04/specification.md`. Includes numbered Functional Requirements (FR-01+), Business Rules (BR-01+), the complete Ticket status-transition matrix (BR-09), the IT-Staff resolution gate (BR-08), dashboard calculation rules (BR-10), two justified database-design decisions, Acceptance Criteria (AC-01+), and the Product Definition of Done. The specification was committed (PR #56) before the implementation pull requests, evidencing specification-before-code.

Answer Part 3: Test DD and Traceability

Rendered: `docs/lab-04/tests.md` — planned tests and a traceability matrix mapping each Acceptance Criterion to its automated test file. Passing suites on `main`:

```
server:  Test Files 13 passed (13)    Tests 46 passed (46)
client:  Test Files 10 passed (10)    Tests 18 passed | 2 todo (20)
tsc --noEmit: clean        vite build: OK
```

Test files: `server/tests/lab-04/actions-taken.api.test.ts`, `ticket-workflow.api.test.ts`, `requester-dashboard.api.test.ts`, `staff-dashboard.api.test.ts`; `client/tests/lab-04/ActionsTaken.test.tsx`, `RequesterDashboard.test.tsx`, `StaffDashboard.test.tsx`, `Responsive.test.tsx`.

[Passing Test Output Screenshot — server and client suites]

Answer Part 4: AI Use & Reflection

Rendered: `docs/lab-04/ai-use.md` — the LLM tools used, 8 selected key engineering prompts with their verifications, and a brief reflection on specification-agent and coding-agent use.

Answer Part 5: Working IT Staff Dashboard UI

Backend endpoint `GET /api/dashboard/staff` returns unassigned, owned-by-me, by-status, by-priority, and recently-updated metrics computed from authoritative data (verified in `staff-dashboard.api.test.ts`). The UI (`client/src/Dashboard.tsx`) is reached from the Dashboard navigation item; metric cards drill through to the Staff Queue.

[Screenshot: IT Staff Dashboard — metrics and recent tickets]

Answer Part 6: Working Actions Taken UI

Endpoints `GET/POST/PATCH /api/tickets/:id/actions` enforce authorization on the backend (IT Staff/Administrator write; Requester reads only owned tickets) and require a follow-up note when follow-up is flagged. The UI (`client/src/ActionsTaken.tsx`) on the Ticket Detail screen supports a list (oldest-first), create and edit modes, a duplicate-submit guard, and read-only display for Requesters.

[Screenshot: Ticket Detail with multiple Actions Taken and the create form]

Answer Part 7: Working Ticket Workflow

The backend enforces the full eight-status transition matrix (New, Open, In Progress, Waiting for Requester, Resolved, Closed, Reopened, Cancelled). Illegal transitions return 400; stale updates return 409; a Requester "appears resolved" indication is advisory and never sets the ticket to Resolved. Verified in `ticket-workflow.api.test.ts`.

[Screenshot: status control showing only permitted transitions]

Answer Part 8: Working Requester Dashboard and Final Regression

The Requester dashboard (`GET /api/dashboard/requester`) is owner-scoped: total open, waiting-for-me, recently updated, and recently resolved — ownership enforced on the backend (verified in `requester-dashboard.api.test.ts`). All Lab 1–3 suites continue to pass on `main`, confirming preserved authentication, My Tickets, Ticket Detail, attachments, comments, IT Staff functions, and administrator user management.

[Screenshot: Requester Dashboard and representative regression screens]

Answer Part 9: Zen Green UI and Responsive Evidence

Rendered: `docs/lab-04/ui-spec.md`. Root-cause fix for the mobile defect: the application previously served a stale compiled `App.js`, so edits to `App.tsx` never shipped; `main.tsx` and the tests now load `App.tsx`, and `tsconfig noEmit:true` prevents re-emitting the compiled files. Responsive styles add breakpoints at 768px and 1024px, horizontally scrollable table wrappers, full-width inputs (16px to prevent iOS zoom), a wrapping navigation bar, and visible `:focus-visible` outlines; no horizontal page scroll remains.

[Screenshots: desktop, tablet, and mobile views of the major Lab 4 screens]

The submitted repository and final `main` branch remain the source of truth. Screenshot placeholders above are to be filled from the running application before final submission.